Minimum Bactericidal Concentration (MBC) of Papain

MATERIALS AND METHODS

1) Treatment of bacterial samples with papain

Various concentrations of 0.1%, 0.5%, 1.0%, 2.5% w/v papain in deionised water, as well as phosphate buffered saline (PBS) for the control experiment, were prepared. 10ml of liquid cultures of *Bacillus cereus, Bacillus subtilis, Staphylococcus epidermidis, Streptococcus mutans and Escherichia coli* were obtained and vortexed to suspend the bacterial cells and ensure even distribution. 0.2ml of each bacterial suspension was added to 0.8ml of each concentration of papain. Similarly, 0.2ml of each bacterial suspension was added to 0.8ml of PBS as control. All samples were then incubated at 37°C for 24 hours.

2) Observation of bactericidal effect

All bacterial samples were centrifuged at 13 300 rotations per minute (rpm) for 5 minutes and the supernatant was then removed. An 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT) essay was then conducted. 0.5ml of MTT was then added to the remaining bacterial pellet and then incubated at 37°C for 3 hours. After 3 hours, the samples were observed for the formation of purple formazan crystals, which indicated the presence of living bacterial cells. Samples that showed no crystal formation indicated bactericidal effect with insignificant or no presence of living bacterial cells.

3) Determining MBC

From the results in (2), concentrations of papain were narrowed down to 0.1% divisions and respectively diluted from the originally prepared samples. Treatment procedure (1) and observation procedure (2) were repeated using the corresponding papain concentrations.

4) Observation of bactericidal effect in ambiguous situations

When anomalous effects were observed, whereby MTT assay indicated bactericidal effects in lower papain concentrations but not in higher concentrations for a specific bacteria type, the treatment procedure (1) was repeated. 0.2ml of each bacterial sample was then added and spread to a Luria-Bertani (LB) agar plate. The bacteria plates were then incubated at 37°C for 24 hours and bacterial growth was observed afterward.

RESULTS

Bacteria species	MBC of papain / %
Bacillus cereus	0.6
Bacillus subtilis	0.6
Staphylococcus epidermidis	0.2
Streptococcus mutans	0.2
Escherichia coli	> 2.5

Table 1: MBC of papain

Protoplast Isolation, Regeneration, and PEG-Induced Fusion between *Pleurotus Ostreatus* and *Ganoderma Lingzhi* to create heat-tolerant hybrid strains

MATERIALS AND METHODS

1) Identification of suitable digestion mixture for protoplast isolation

0.6M KCl w/v in deionised water was prepared and autoclaved to sterilise. 10mg/ml solutions of cellulase, pectinase and lysozyme were prepared using 0.6M KCl as solvent and then filter-sterilised. 4 different tubes of digestion mixture were prepared that were 3ml of cellulase, 3ml of pectinase, 3ml of lysozyme and a mixture of 1ml of each enzyme were prepared. A cut piece of solid mycelia from *Ganoderma Lingzhi* was added to each digestion mixture, and then incubated in the orbital shaker at 100 rotations per minute (rpm) at 30°C for 3 hours. Each tube was vortexed and then had 100µl of liquid extracted for observation under a microscope every hour. After 3 hours, 5ml of Sabouraud broth was added to every tube and subsequently incubated at room temperature for 72 hours.

2) Protoplast isolation and regeneration

When the most suitable enzyme was identified as cellulase, two tubes of 1ml of cellulase were prepared. Cut pieces of solid mycelia from *Ganoderma Lingzhi* and *Pleurotus Ostreatus* each were added to the tubes containing cellulase and the mixture was then incubated in the orbital shaker at 100 rpm at 30°C for 3 hours. Afterwards, the liquid from the tubes were completely extracted and filtered through a cell strainer to remove undigested debris. The filtrate was then centrifuged at 13 300 rpm for 5 minutes and the supernatant was removed. The remaining pellet was then resuspended with 1ml of 0.6M KCl osmotic stabiliser. 0.1ml of each resuspended protoplast was then added to 3ml of Sabouraud broth and incubated at room temperature for 72 hours.

3) Protoplast inactivation

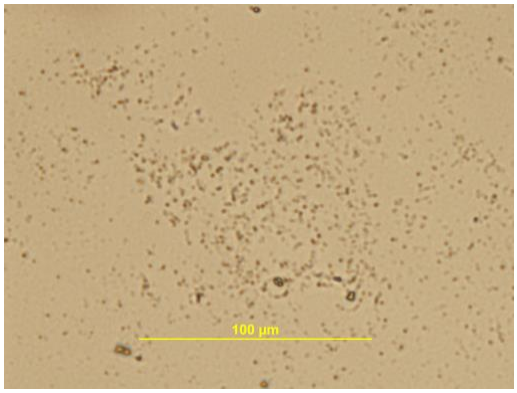
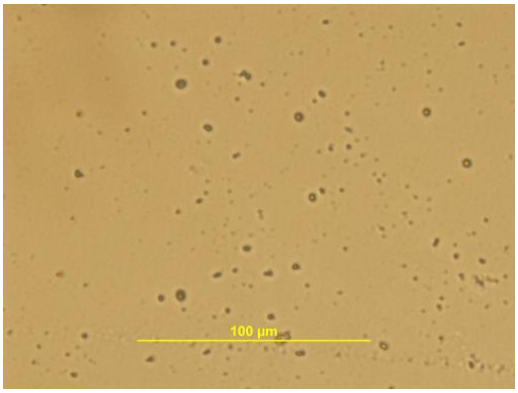
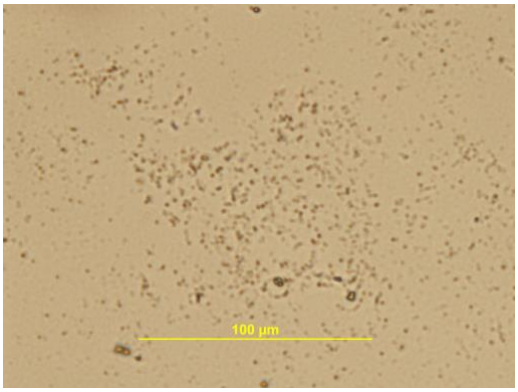
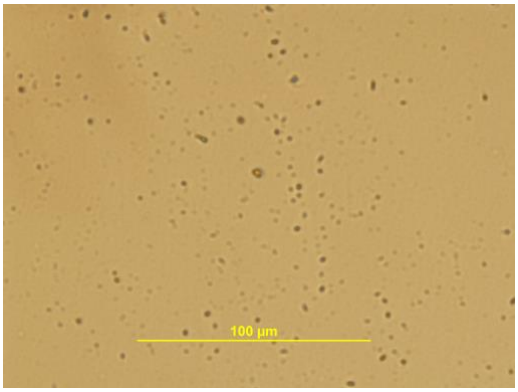
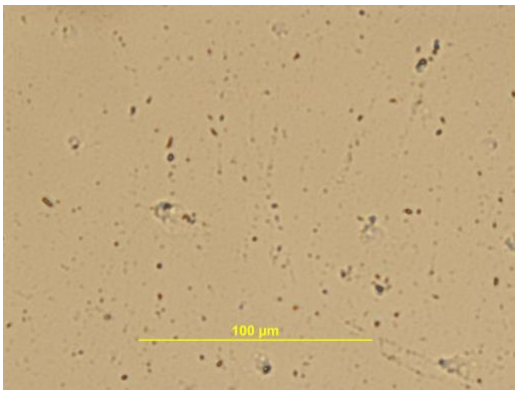
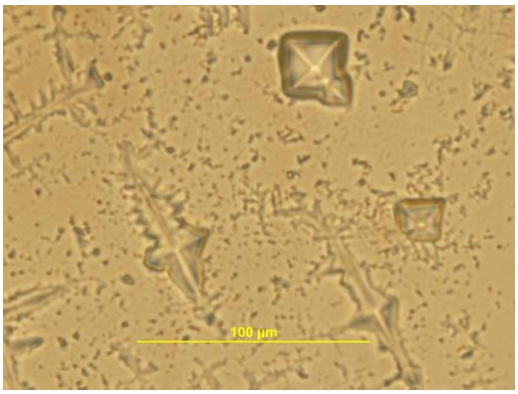
The remaining *Pleurotus Ostreatus* protoplasts were then incubated in a water bath at 50°C for 30 minutes to inactivate the *Pleurotus Ostreatus* protoplasts. Afterwards, 0.1ml of inactivated protoplast was added to 3ml of Sabouraud broth and incubated at room temperature for 72 hours.

4) Protoplast fusion and growth

40% w/v PEG of molecular weight 6000 and 0.04M $CaCl_2$ was prepared. 0.5ml of each protoplast suspension was then mixed together in a tube and 1ml of PEG was then added to the same tube. The tube was then incubated in the orbital shaker at 100 rpm at 30°C for 30 minutes. Afterwards, the tube was centrifuged at 13 300 rpm for 5 minutes and the supernatant was removed. The remaining pellet was then transferred to and grown on a Sabouraud agar plate and incubated at 37°C for 72 hours. One *Ganoderma Lingzhi* and *Pleurotus Ostreatus* plate each was also subcultured at incubated at 37°C for 72 hours to ensure that only *Pleurotus Ostreatus* and the fusant would grow in high heat conditions.

RESULTS

Table 1: Effect of various digestion mixtures and incubation times on protoplast isolation

Enzyme	1 hour	3 hours
Pectinase		
Cellulase		
Lysozyme		

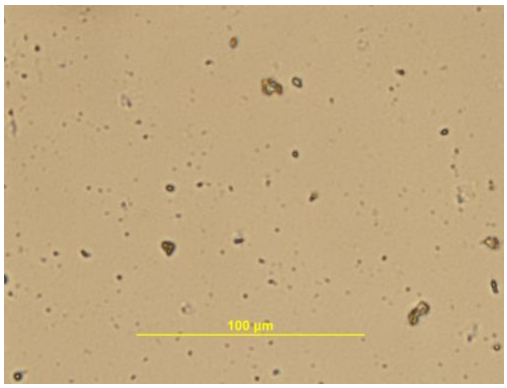
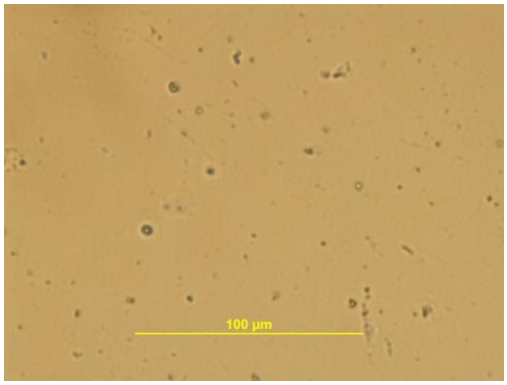
Mixed		

Table 2: Protoplast regeneration of *Ganoderma Lingzhi* after isolation with various enzymes (see Appendix, Figure A1)

Enzyme used in protoplast isolation	**Successful regeneration?**
Pectinase	Yes
Cellulase	Yes
Lysozyme	Yes
Mixed	Yes

Table 3: Protoplast regeneration and inactivation before fusion

Type	**Successful regeneration?**	**Absence of growth after inactivation?**
Pleurotus Ostreatus	No	Yes
Ganoderma Lingzhi	Yes	-

Table 4: Growth of parent strains and fusant in 37°C (see Appendix, Figure A2)

Type	**Successful growth?**
Pleurotus Ostreatus	Yes
Ganoderma Lingzhi	No
Fusant	No

DISCUSSION

Protoplast isolation using the various digestion mixtures was generally successful with a good protoplast yield for each digestion mixture. At 1 hour, some protoplasts were formed but the

yield was lower than at 3 hours. Apart from that, certain digestion mixtures also showed protoplasts that had already lysed, indicating the possibility that the osmotic stabiliser chosen was poor and the excessive osmotic pressure led to the protoplasts breaking apart.

However, protoplast yields were much higher for all digestion mixtures at 3 hours, and few lysed protoplasts were observed. The protoplast yield was highest for cellulase followed closely by pectinase, whereas the mixed digestion mixture had the lowest yield. Multiple crystals were observed for the digestion mixture containing lysozyme at 3 hours which was likely due to incomplete dissolving of lysozyme in the solution. Hence, cellulase was selected as the most suitable enzyme for use in the digestion mixture for protoplast isolation.

Protoplasts in all digestion mixtures underwent successful regeneration after Sabouraud broth was added and the mixtures were incubated, indicating that live protoplasts were obtained in the protoplast isolation phase. Therefore, no changes were made to the KCl osmotic stabiliser chosen.

The protoplast fusion process was unsuccessful as no growth was observed on the fusant plates. This is likely due to the fact that *Pleurotus Ostreatus* protoplasts were not successfully obtained as no growth was observed in the regeneration phase. As the determination of suitable digestion enzymes was conducted on *Ganoderma Lingzhi*, there may be a need to explore the suitability of various enzymes in the protoplast isolation of *Pleurotus Ostreatus*. A liquid culture of *Pleurotus Ostreatus* may also be explored instead of using a solid mycelial mass for the protoplast isolation.

As only *Pleurotus Ostreatus* and not *Ganoderma Lingzhi* grew in 37°C, this confirms that any growth of fusant at 37°C would only be possible if fusion was successful as *Pleurotus Ostreatus* was inactivated prior to fusion.

CONCLUSION

Protoplasts were successfully isolated and regenerated for *Ganoderma Lingzhi*, but not for *Pleurotus Ostreatus*. Hence, protoplast fusion was also unsuccessful and no hybrid strains were created. Other enzymes or mycelia sources can be considered for successful protoplast isolation of *Pleurotus Ostreatus* prior to fusion.

APPENDIX

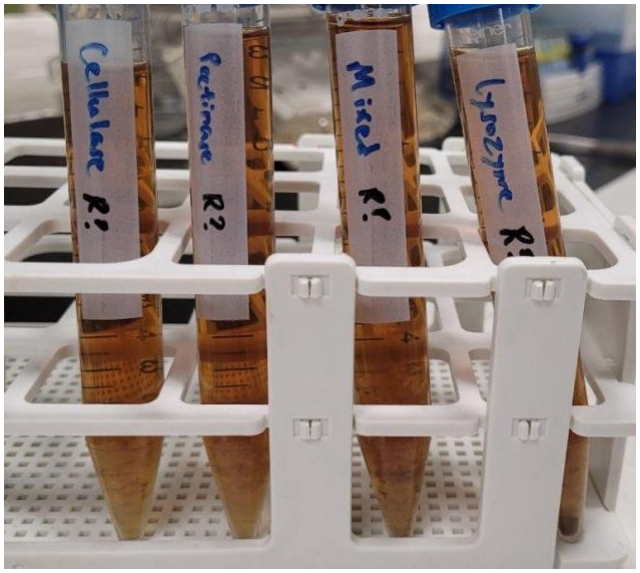


Figure A1: Protoplast regeneration of *Ganoderma Lingzhi* after isolation with various enzymes

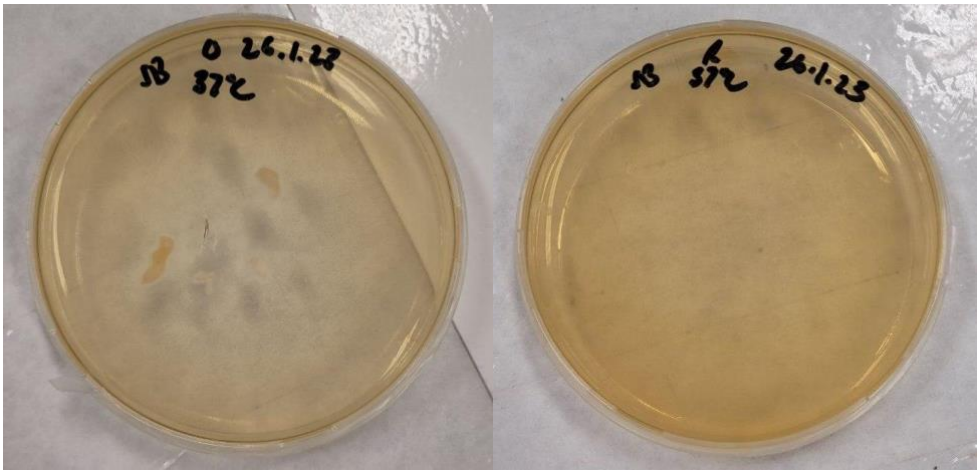


Figure A2: Growth of *Pleurotus Ostreatus* (left) and *Ganoderma Lingzhi* (right) in 37°C

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Recovery of Enzymes of Spent Mushroom Compost from *Pleurotus Ostreatus* and subsequent applications in the removal of ultraviolet light filters from sunblock

MATERIALS AND METHODS

1) Recovery of enzymes of spent mushroom compost (SMC)

A mass of sugarcane bagasse SMC from *Pleurotus Ostreatus* was obtained. 50mM of sodium citrate ($Na_3C_6H_5O_7$) solution was prepared. 2 tubes, each containing 33ml of sodium citrate solution added to 10g of SMC, were prepared, alongside one tube containing 33ml of deionised water and 10g of SMC and a metal spoon was used to break up the SMC into smaller pieces. All mixtures were incubated in the orbital shaker at 200 rotations per minute (rpm) at room temperature for 24 hours. After 6 hours, one SMC-sodium citrate mixture was removed. The mixture was centrifuged at 9000 rpm for 15 minutes and the supernatant containing the enzymes was extracted and stored at -18°C. After 24 hours, all remaining mixtures were removed and similar preparation processes and storage conditions were repeated.

2) Preparation of Bradford Assay Stand Curve

Multiple solutions of differing concentrations of bovine serum albumin (BSA), 10µg/ml, 20µg/ml, 30µg/ml, 40µg/ml and 50µg/ml of BSA, were prepared, each at volumes of 0.2ml using deionised water as solvent. To each BSA solution, 1ml of Bradford's reagent was added. Similarly, to 0.2ml of deionised water, 1ml of Bradford's reagent was added to prepare a blank. All mixtures were then incubated for 10 minutes at room temperature. The absorbance of all reaction mixtures were then obtained at wavelength of 595nm, with the deionised water and Bradford's reagent mixture used as blank. A graph of absorbance of solution OD_{595} against concentration of BSA/µg ml^{-1} was plotted and a best fit line was drawn to obtain the standard curve.

3) Bradford Assay

To 0.1ml of each SMC extract, 0.1ml of deionised water was added for dilution purposes, followed by 1ml of Bradford's reagent. Similarly, a blank was prepared with 0.2ml of deionised water and 1ml of Bradford's reagent. All mixtures were incubated for 10 minutes at room temperature and subsequently, the absorbance of all mixtures were obtained at wavelength of 595nm with deionised water and Bradford's reagent mixture used as blank. The corresponding protein concentration in the SMC extract was then determined by calculation from the standard curve obtained previously.

4) Preparation of sunblock solution of ultraviolet (UV) filters

To 200ml of deionised water, a minimal amount of sunblock was added such that the water continued to appear visually clear. The mixture was stirred thoroughly to dissolve as much sunblock as possible and was then centrifuged at 11 000 rpm for 10 minutes. The supernatant

was extracted and then filtered through a 0.45µm filter to remove all remaining undissolved particles. The resultant sunblock solution was then stored at room temperature in the dark.

5) Determination of effect of UV light on UV filters in sunblock

A small volume of filtered sunblock was poured onto a petri dish and treated with UV light for 30 minutes. Subsequently, 1ml of deionised water, 1ml of untreated sunblock and 1ml of UV-treated sunblock were added to separate cuvettes. Using 1ml of deionised water as blank, the percentage transmission of light (% transmission of light) through each solution was measured using the spectrophotometer across a spectrum of light wavelengths ranging from 280nm to 400nm.

6) Immobilisation of enzymes of SMC on silica gel

A sufficient amount of the SMC extract (sodium citrate, 24 hour incubated) was added to completely cover a 5g mass of silica gel. The mixture was then incubated for 24 hours. A Büchner funnel was set up with 2 pieces of damp filter paper lining the funnel. The wet silica gel mixture was poured into the funnel and the tube was washed with deionised water to ensure no residue silica gel was left. The silica gel was then left to dry in the Büchner funnel for 3 minutes before the enzyme-embedded silica gel was removed and stored.

7) Immobilisation of enzymes of SMC on alginate-silica beads

A 5g mass of silica gel and 50ml of 3% sodium alginate solution were added to a beaker and stirred to keep the silica gel in suspension. A separate beaker of 10% calcium chloride ($CaCl_2$) solution was also prepared. Using a micropipette and a cut micropipette tip, the sodium alginate-silica gel suspension was added dropwise to the $CaCl_2$ solution to form the alginate-silica beads. The beads were then left to cure in the $CaCl_2$ solution for 10 minutes before they were removed from the solution and left to dry in the oven at 55°C for 24 hours. The beads were separated into two portions. To one portion of the beads, a sufficient amount of the SMC extract (sodium citrate, 24 hour incubated) was added to completely cover the beads. To the other portion, a sufficient amount of deionised water was added to completely cover the beads. Both mixtures were then incubated for 24 hours before. A Büchner funnel was set up with 2 pieces of damp filter paper lining the funnel. One portion of beads was poured into the funnel and left to dry in the Büchner funnel for 3 minutes before the alginate-silica beads were stored. The process was repeated separately for the other portion of beads.

8) Treatment of UV filters in sunblock with enzymes of SMC

The following tubes were prepared:

1. 2ml deionised water + 5ml sunblock solution [Control]
2. 2ml SMC extract (sodium citrate, 24 hour incubated) + 5ml sunblock solution
3. 2g clean silica gel (with no embedded enzymes) + 5ml sunblock solution
4. 2g enzyme-embedded silica gel + 5ml sunblock solution

5. 2g clean alginate-silica beads (with no embedded enzymes) + 5ml sunblock solution
6. 2g enzyme-embedded alginate-silica beads + 5ml sunblock solution
7. 2g SMC + 5ml sunblock solution

All mixtures were then thoroughly mixed together. All the tubes were then incubated in the dark for up to 5 days and 1ml of solution extracted for testing periodically.

9A) Determination of effect of enzymes of SMC on UV filters in sunblock

Tubes 3, 4, 5, 6 and 7 were centrifuged at 11 000 rpm for 10 minutes. 1ml of the supernatant in each tube was then extracted and added to a cuvette. 1ml of each solution in the remaining tubes was also extracted and added to a cuvette. 1ml of deionised water was also added to another cuvette as a blank. Using 1ml of deionised water as blank, the % transmission of light through each solution was measured using the spectrophotometer across a spectrum of light wavelengths ranging from 280nm to 400nm. All samples were then returned to their respective tubes to maintain similar volumes.

9B) Determination of effect of enzymes of SMC on UV filters in sunblock

Tube 2 was centrifuged at 11 000 rpm for 10 minutes. 1ml of supernatant in tube 2 was then extracted and added to a cuvette. 1ml of solution in tube 1 was also extracted and added to a cuvette. Another 2 cuvettes were prepared, one containing 1ml of deionised water and one containing 1ml of deionised water and 0.4ml of SMC (extract sodium citrate, 24 hour incubated). Using deionised water as blank, the % transmission of light through solution from tube 1 was measured using the spectrophotometer across a spectrum of light wavelengths ranging from 280nm to 400nm. Using deionised water-SMC extract mixture as blank, the % transmission of light through solution from tube 2 was measured using the same process. All samples were then returned to their respective tubes to maintain similar volumes.

RESULTS

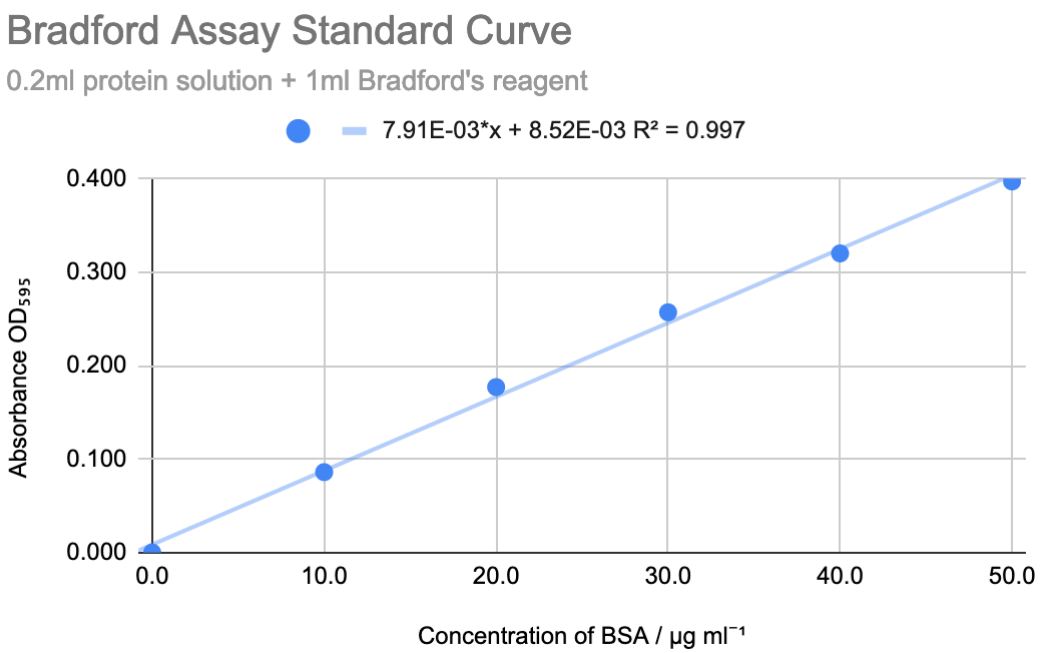

Figure 1: Bradford Assay Standard Curve

Extraction medium	**Incubation time**	**Protein concentration / µg ml^{-1}**
Sodium citrate	6 hours	81.54
Sodium citrate	24 hours	87.61
Deionised water	24 hours	82.30

Table 1: Protein concentration of different SMC extraction types

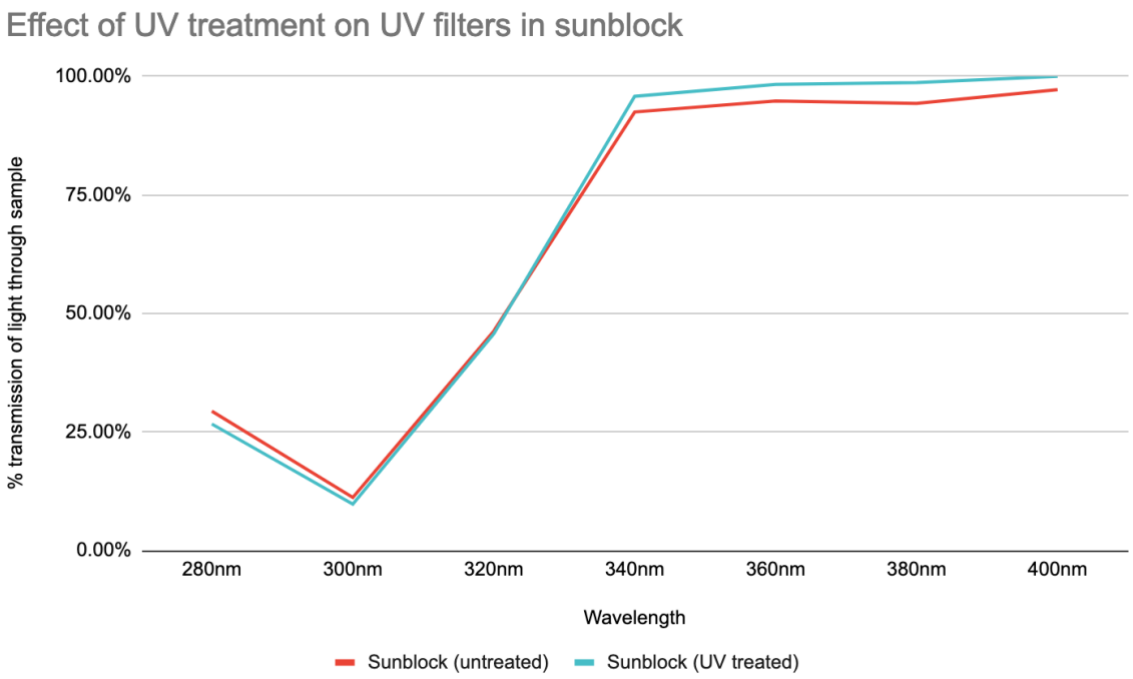

Figure 2: Effect of UV treatment on UV filters in sunblock

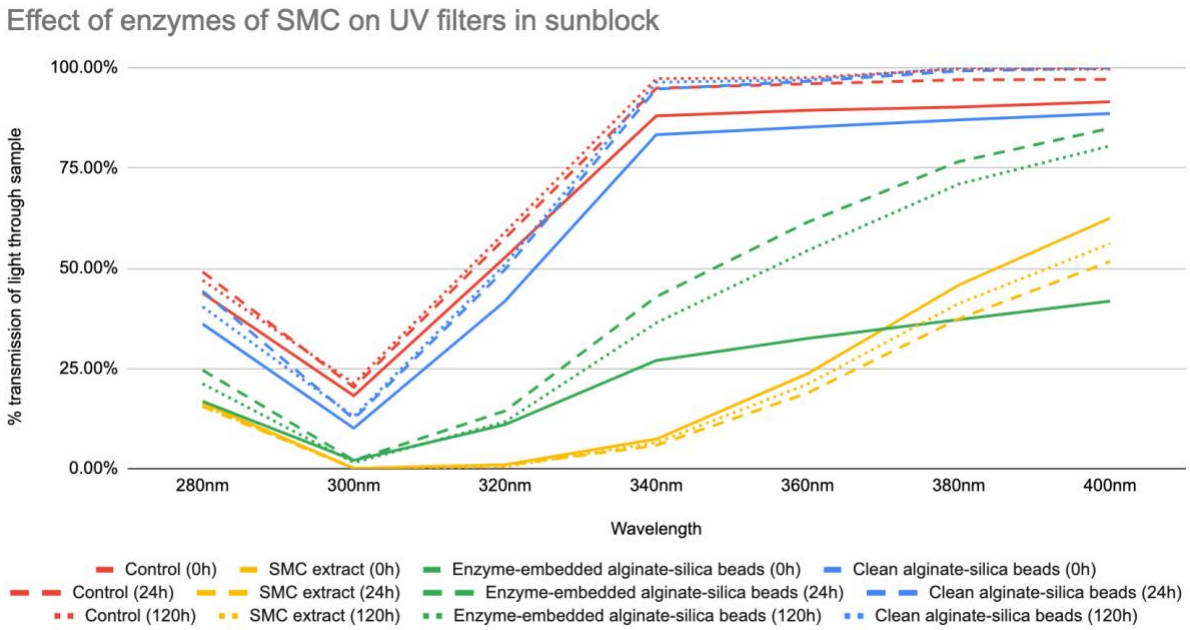

Figure 3: Effect of enzymes of SMC on UV filters in sunblock (procedure 9A)
* *no data for tubes 3, 4 and 7 is available due to reasons outlined in discussion*

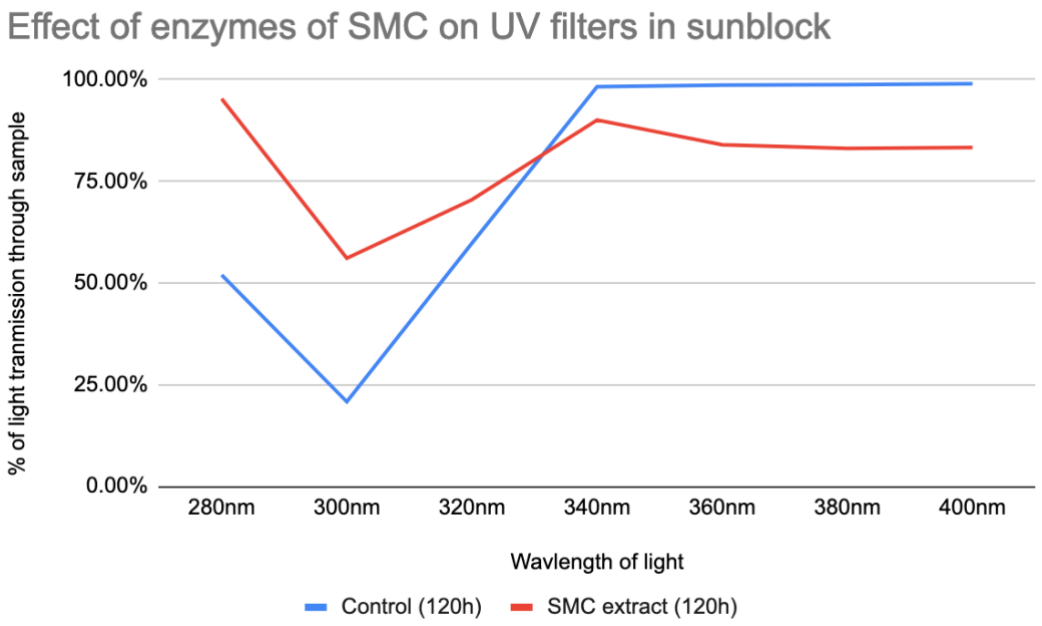

Figure 4: Effect of enzymes of SMC on UV filters in sunblock (procedure 9B)

DISCUSSION

The extraction of enzymes of SMC was considered to be highly effective as a significant protein concentration was present in the final SMC extract. However, as the Bradford assay is a generic test for protein concentration, it was not possible to determine if the proteins present in the SMC extract are enzymes or other proteins. Further enzyme-specific tests on starch or cellulose may be necessary to more accurately determine the enzymatic activity of the SMC extract. Moreover, as there was no control setup for the extraction of enzymes of SMC, the source of proteins or enzymes is also ambiguous and may be from the sugarcane bagasse itself or from *Pleurotus Ostreatus*. It can also be noted that the different extraction media and extraction incubation time was shown to have little effect on the final protein concentration in the SMC extract.

Despite the concentration of sunblock in solution being very low, the sunblock solution itself is still able to block out more than 90% of UV light at the wavelength of 300nm and also has similar blocking properties at the other UV wavelengths. Furthermore, after UV treatment of the UV filters in sunblock, there was minimal degradation of the UV filters. The % of transmission of light through both the treated and untreated samples was similar throughout, and the UV treated sample had a slightly lower % transmission of light through it at wavelengths less than 320nm, a result likely due to the increase in concentration of UV filters due to evaporation of water from the petri dish. This hence shows the potency of sunblock being washed into large water bodies and its long-lasting environmental impacts as it is not broken down over time.

In general, the results from treatment of sunblock with enzymes of SMC are inconclusive and further data will need to be gathered to ascertain the effectiveness of such enzymes. Initially, silica gel was used to immobilise the enzymes of SMC to react with the sunblock solution. However, as the initial sunblock solution concentration was excessively high, transmission of light through the solution remained at 0.0% for the UV-B wavelength (280nm to 315nm) which indicates excessive concentrations of UV filters which block light at the UV-B wavelength. At the UV-A wavelengths (315nm to 400nm), improvements in transmission observed for both silica gel samples with or without enzymes were similar in magnitude and hence the experiment was redesigned and sunblock solution greatly diluted.

While trials were also run using the SMC added directly to the sunblock solution, this trial faced a great number of difficulties, including the SMC absorbing the solution and causing a significant decrease in volume of solution, difficulty in separating supernatant from SMC as well as presence of proteins and other possible molecules released by SMC into the sunblock solution that hinders the transmission of UV light through it. However, direct application of SMC with living *Pleurotus Ostreatus* mycelium is another alternative that should be explored as the potential to continuously secrete enzymes may have significant potential in removing UV filters from sunblock.

Where the SMC extract was used, presence of small particles as well as the ability of protein to absorb wavelengths of light similar to the UV-B wavelength caused transmission to be much lower than expected at the UV-B wavelength. Although alginate-silica beads were devised to fix the enzymes in place and minimise this issue, the solution turned cloudy after alginate-silica beads were added (see Appendix, figure A1) suggesting the possibility that the silica gel or enzymes had leaked out into the solution and hence caused data to be unreliable. Centrifugation was explored as an option to improve this situation but was not highly successful. In Figure 3, all data points for sunblock solution mixed with another solution intended to remove UV filters ended up with lower % of light transmission through sample, indicating poor experiment design contributing to unreliable results.

Procedure 9B aimed to fix this issue by ensuring all solutions are blanked with their corresponding solution in deionised water instead of sunblock solution. Figure 4 suggests that the mushroom extract was able to reduce the concentration of UV filters that blocked light in the UV-B wavelengths, but instead still increased the concentration of UV filters that blocked light in the UV-A wavelength after correct blanking. Furthermore, while not reflected in Figure 4, the shape of the % of light transmission through sample of Sunblock + SMC extract was not smooth nor continuous, but instead spiked at various wavelengths (see Appendix, figure A2), although the general shape of the graph is still the same. Due to time constraints, no accurately blanked data points could be obtained for the alginate-silica beads or silica gel. Apart from that, data obtained following procedure 9B was also only taken at 1 time, and not over a period of times.

From Figure 4, enzymes of SMC do show the potential in breaking down and removing UV filters from sunblock solution. However, alternatives need to be explored in ensuring proper immobilisation of the enzymes to fix the enzymes in place in a way that would not cause proteins or other solid particulates to be released into the sunblock solution that could potentially increase the concentration of UV filters. One such possible solution is to prepare alginate-silica beads together with the SMC extract instead of incubating the beads in SMC extract afterwards. Ideally, the beads should then be incubated in deionised water for 24 hours to determine if the solution turns cloudy or if any small solid particles are seen to be released.

CONCLUSION

In conclusion, extraction of enzymes from SMC was highly successful with a good yield of high protein concentration, but further work is required to determine enzymatic activity of extract and determine if the source of enzymes are sugarcane bagasse or from *Pleurotus Ostreatus* in the SMC.

The application of enzymes of SMC in the removal of UV filters in sunblock showed positive results in preliminary trials indicating potential ability in mycoremediation, but was obfuscated by multiple factors causing the result to be considered inconclusive. Further work is required to properly immobilise enzymes of SMC on a surface and to explore using SMC with living *Pleurotus Ostreatus* mycelium to more effectively remove UV filters in sunblock.

APPENDIX

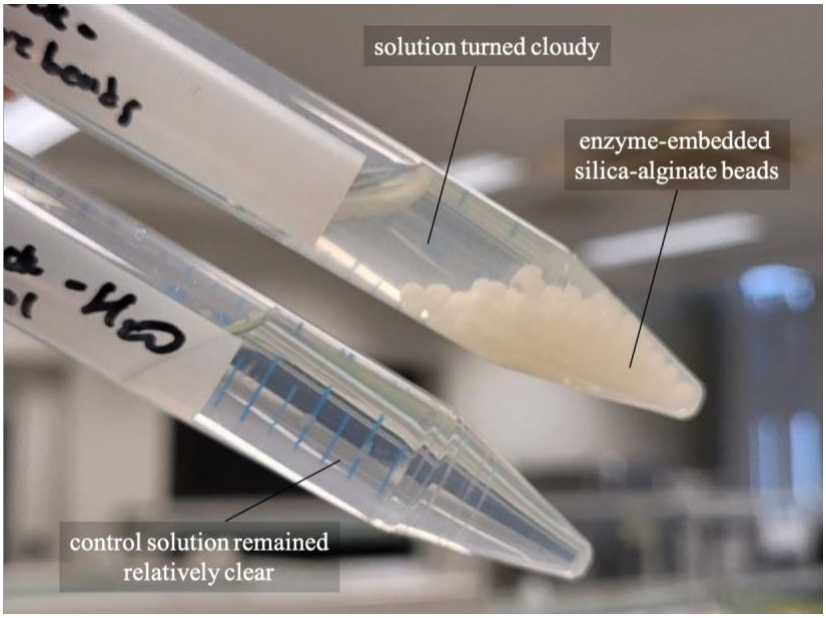

Figure A1: Cloudiness of solution with alginate-silica beads against control

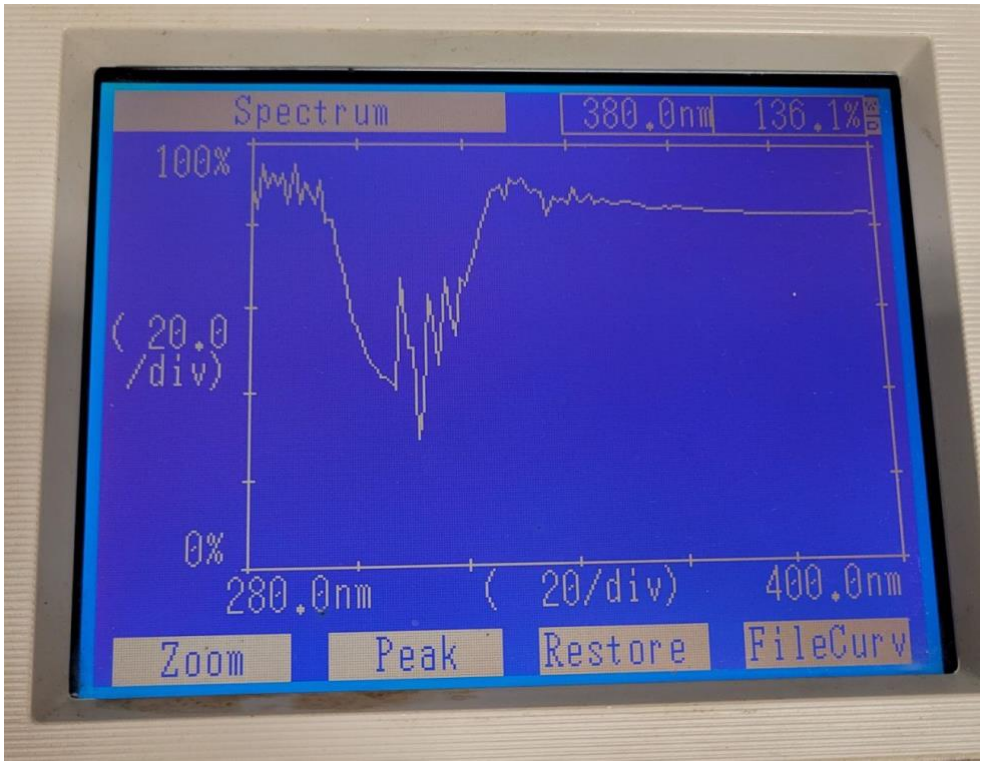

Figure A2: Spectrum analysis of Sunblock + SMC extract graph (procedure 9B)

Sample	% transmission at wavelengths / %						
	280nm	**300nm**	**320nm**	**340nm**	**360nm**	**380nm**	**400nm**
Sunblock (untreated)	29.40%	11.20%	46.30%	92.50%	94.80%	94.30%	97.20%
Sunblock (UV treated)	26.70%	9.80%	45.70%	95.80%	98.30%	98.70%	100.00%

Table A1: Data for Figure 2

Sample	% transmission at wavelengths / %						
	280nm	**300nm**	**320nm**	**340nm**	**360nm**	**380nm**	**400nm**
Initial (0 hr)							
Control	43.80%	18.20%	52.70%	88.00%	89.40%	90.20%	91.50%
SMC extract	16.40%	0.10%	1.00%	7.40%	23.70%	45.80%	62.50%
Enzyme-embedded alginate-silica beads	16.80%	2.10%	11.00%	27.00%	32.50%	37.20%	41.80%
Clean alginate-silica beads	36.10%	10.10%	41.80%	83.30%	85.20%	87.00%	88.60%
Day 1 (24 hr)							

Control	49.10%	20.30%	57.50%	94.90%	96.00%	97.00%	97.10%
SMC extract	15.50%	0.10%	0.80%	5.80%	18.90%	37.40%	51.70%
Enzyme-embedded alginate-silica beads	24.60%	2.10%	14.40%	42.90%	61.50%	76.60%	84.90%
Clean alginate-silica beads	44.30%	12.40%	49.70%	94.70%	96.60%	99.20%	99.90%
Day 5 (120 hr)							
Control	47.00%	21.30%	59.00%	97.30%	97.50%	99.70%	99.70%
SMC extract	15.70%	0.10%	0.50%	6.50%	21.10%	41.20%	56.10%
Enzyme-embedded alginate-silica beads	21.20%	1.50%	11.70%	36.40%	54.40%	71.00%	80.50%
Clean alginate-silica beads	40.40%	13.00%	51.00%	96.40%	97.00%	100.00%	100.00%

Table A2: Data for Figure 3 (procedure 9A)

Sample	**% transmission at wavelengths / %**						
	280nm	**300nm**	**320nm**	**340nm**	**360nm**	**380nm**	**400nm**
Control (120h)	52.00%	20.80%	59.80%	98.30%	98.70%	98.80%	99.00%
SMC extract (120h)	95.30%	56.10%	70.50%	90.10%	84.00%	83.10%	83.30%

Table A3: Data for Figure 4 (procedure 9B)

REFERENCES

Lim, S. H., Lee, Y. H., & Kang, H. W. (2013). Efficient Recovery of Lignocellulolytic Enzymes of Spent Mushroom Compost from Oyster Mushrooms, *Pleurotus* spp., and Potential Use in Dye Decolorization. *Mycobiology*, *41*(4), 214–220. https://doi.org/10.5941/myco.2013.41.4.214

Yang, C. W., Tu, P. H., Tso, W. Y., & Chang, B. V. (2021). Removal of Organic UV Filters Using Enzymes in Spent Mushroom Composts from Fungicultures. *Applied Sciences*, *11*(9), 3932. https://doi.org/10.3390/app11093932

Removal of Heavy Metal Ions Using Modified Alginate Beads
**Only procedure and result available*

MATERIALS AND METHODS

1) Preparation of standard curves
A standard curve for Cu^{2+} concentration with Zincon was obtained from online sources. 1ml of various concentrations of bromophenol blue, namely 2 ppm, 4 ppm, 6ppm, 8 ppm and 10 ppm were prepared. 1 ml of deionised water was also prepared as a blank. The absorbance of all samples was then obtained at a wavelength of 590 nm using deionised water as blank. A graph of absorbance of solution OD_{590} against concentration of bromophenol blue/ppm was plotted and a best fit line was drawn to obtain the standard curve. The entire procedure was then repeated using methylene blue with identical concentrations of dye and absorbance obtained at the same wavelength.

2) Preparation of eggshells and beads
A mass of eggshells was obtained and washed to remove any leftover egg white or yolk. The eggshells were then dried in an oven at 70°C for 30 minutes. The thin white membrane on the inner surface of the eggshells was then removed from half of the eggshells, forming one portion of eggshells without membrane (ES-M) and one portion of eggshells with membrane (ES+M). Using a pestle and mortar, the eggshells were grinded into a fine powder. In separate beakers, 30ml of 3% sodium alginate solution was added to 5g of each type of eggshell. A third solution containing 30ml of 3% sodium alginate solution with no eggshells was also prepared. The solutions were mixed and then added dropwise using a cut micropipette tip to 5% calcium chloride solution and then incubated for 10 minutes. The beads were then removed from the calcium chloride solution.

3) Preparation of solutions containing pollutants
4 tubes containing 40 ml of 5 ppm Cu^{2+} solution each was prepared. Another 4 tubes containing 40 ml of 10 ppm bromophenol blue solution each was prepared. Another 4 tubes containing 40 ml of 10 ppm methylene blue solution each was prepared. 1 ml of solution from each tube was then extracted and added to a cuvette. Zincon was used for the spectrophotometric determination of Cu^{2+} and Zn^{2+}. To the 1 ml extracted Cu^{2+} solutions, 0.1ml of 1mM stock Zincon solution was added to each solution. The absorbance of Cu^{2+} with Zincon was read at 618nm using 1 ml deionised water + ??ml Zincon as blank and then compared against the standard curve to determine initial concentration of Cu^{2+} in each tube. The absorbance of dye solutions was read at 590 nm using 1 ml deionised water as blank and then compared against the standard curve to determine initial concentration of dye in each tube.

4) Addition of eggshells and orange peel to solutions containing pollutants

2.4g each of cut orange peel, alginate beads, alginate beads (ES+M), and alginate beads (ES-M) were added to each tube containing the Cu^{2+} solution. The same masses were used and the procedure was repeated for the bromophenol blue and methylene blue tubes. 2.4g of cut orange peel was also added to 39 ml of deionised water in another tube. All tubes were then incubated in the orbital shaker at 150 rotations per minute (rpm).

5) Measuring concentration of pollutants

At 2 and 24 hours, 1 ml of solution is extracted from each tube. To the Cu^{2+} solutions, ??ml of Zincon was added to each solution. 1ml orange peel-water solution + ??ml Zincon was prepared as blank for the Cu^{2+} solution incubated with orange peel, whereas 1 ml deionised water + ??ml Zincon was prepared as blank for the other Cu^{2+} solutions. The absorbance of Cu^{2+} with Zincon was read at 618nm using the corresponding blanks and compared against the standard curve to determine the concentration. Similarly, 1ml orange peel-water solution was extracted as blank for dye solutions incubated with orange peel, whereas 1 ml deionised water was used as blank for the other dye solutions. The absorbance of dye solutions was read at 590 nm using the corresponding blanks and then compared against the standard curve to determine initial concentration of each tube. The percentage decrease in pollutant concentration was calculated using the formula:

$$\frac{initial\ concentration - final\ concentration}{initial\ concentration} \times 100\%$$

RESULTS

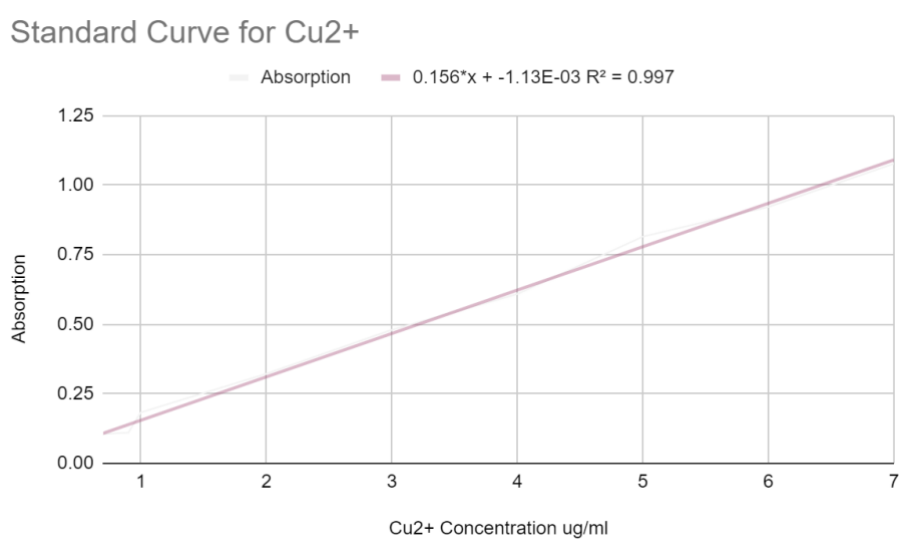

Figure 1: Cu^{2+} with Zincon standard curve

y = 0.0854x +2.86e-03, x=concentration of dye in ppm, 0<ABS<1

Figure 2: Bromophenol blue standard curve equation

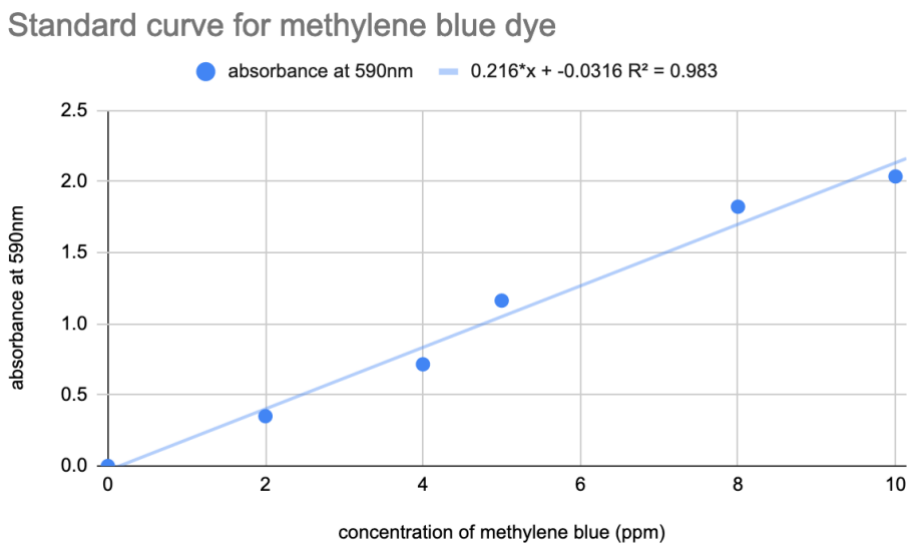

Figure 3: Methylene blue standard curve

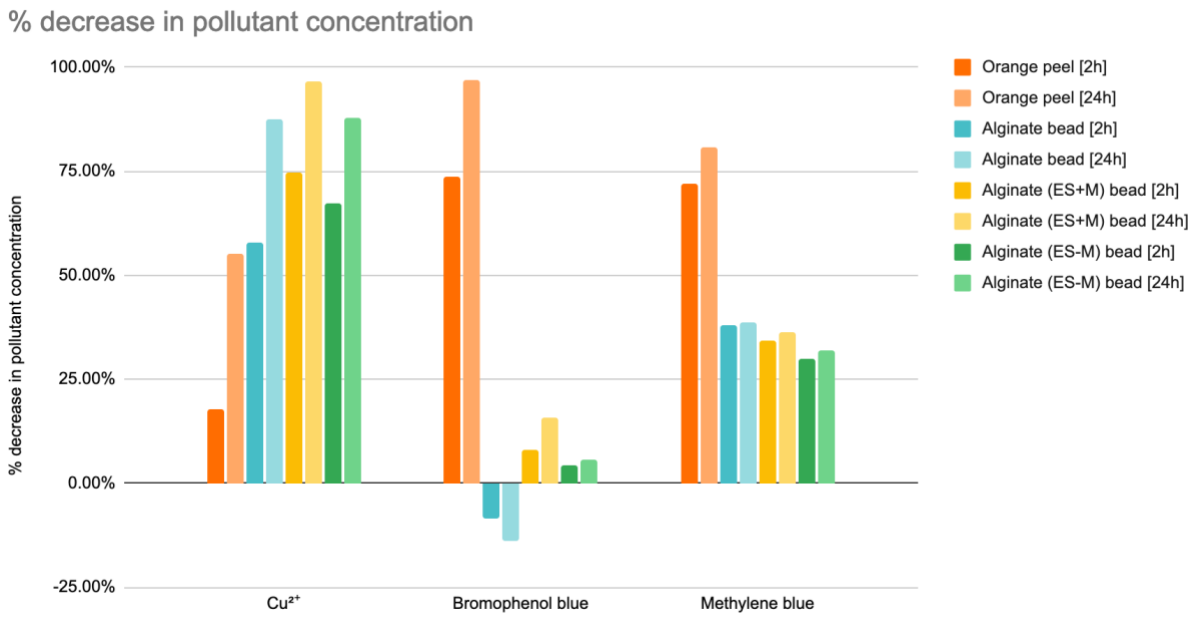

Figure 4: Percentage decrease in pollutant concentration